

Finite-stage purification and local transport

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Scope

Finite Gram matrices and matrix units purify vector states on a chosen finite stage. Corner amplification turns this transport into almost-central unitary paths, with different stage choices for protected-set and unrestricted approximation.

6 direct Cards + 8 reused prerequisites = 14 unique Cards. This count is a selected Card closure, not a source-declaration count.

[Route reading PDF](#) · [Preserved source exploration](#)

Cards in this route

- Key result: Exact vector transport along an almost central unitary path
- Key result: Cross-representation state approximation with a protected finite set

[Read this route with prerequisites](#)

Detailed route conditions

A represented root corner has enough Hilbert-space dimension to realize any finite Gram matrix. Matrix-unit reconstruction then purifies a vector state on a selected finite stage. Supported self-adjoint interpolation and corner amplification turn finite vector transport into unitary paths; the approximate state tests concern every matrix entry, not only diagonal entries.

The protected-set theorem chooses its stage and tolerance before the later state-approximation request. The unrestricted theorem instead purifies on a large stage and transports at stage zero, where the sole unit-vector test is automatic. These are different stages. The fixed-stage abbreviation $e = e_{00}^{(n)}$, the finite-dimensional interpolation space E , and the amplification $L_n(z)$ are local objects, not the later shell links L_i or trace-GNS unitary E .

Dependency-first reading route

Read selected Card prerequisites before their uses. Levels are recomputed from the selected reachability-preserving projection; omitted source helpers remain traceable in the source exploration.

Level 0 (1 Card)

Level 0

The CAR algebra as a completion of finite matrix stages

Names the completed CAR algebra in which finite matrix calculations can be extended by norm approximation.

MathlibAnnex.CStarAlgebra.CAR.Limit

Immediate Card prerequisites: None in this selected scope

Used by in this scope: The product-vector state on the completed CAR algebra · A finite-row average centralizes its matrix stage · The decreasing root projections in the CAR completion · The completed CAR algebra is infinite-dimensional

Open Declaration Card

Read exact source

Level 1 (4 Cards)

Level 1

The product-vector state on the completed CAR algebra

Compatible evaluation at the distinguished matrix coordinate extends continuously to the CAR completion.

MathlibAnnex.CStarAlgebra.CAR.rootState

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: Root compression becomes scalar in norm

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Read exact source

Level 1

The completed CAR algebra is infinite-dimensional

Rules out finite dimension by retaining matrix subalgebras of unbounded dimension.

MathlibAnnex.CStarAlgebra.CAR.not_finiteDimensional

Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages

Used by in this scope: An irreducible CAR representation has no nonzero compact image

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Level 1 The decreasing root projections in the CAR completion The distinguished stage corners give a nested sequence of projections supporting the root state. <code>MathlibAnnex.CStarAlgebra.CAR.rootFlag</code> Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Root compression becomes scalar in norm Open Declaration Card Read exact source
Level 1 A finite-row average centralizes its matrix stage Turns an arbitrary element of the completed CAR algebra into one commuting with a prescribed finite matrix stage. <code>MathlibAnnex.CStarAlgebra.CAR.ofStage_commute_rowAverage</code> Immediate Card prerequisites: The CAR algebra as a completion of finite matrix stages Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path Open Declaration Card Read exact source
Level 2 (1 Card) Level 2 Root compression becomes scalar in norm Extends an exact rank-one compression identity on finite stages to a pointwise norm limit on the completed algebra. <code>MathlibAnnex.CStarAlgebra.CAR.tendsto_norm_compressionError</code> Immediate Card prerequisites: The product-vector state on the completed CAR algebra · The decreasing root projections in the CAR completion Used by in this scope: Simplicity of the completed CAR algebra Open Declaration Card Read exact source
Level 3 (1 Card) Level 3 Simplicity of the completed CAR algebra Every nonzero closed two-sided ideal of the CAR algebra contains the identity. <code>MathlibAnnex.CStarAlgebra.CAR.isSimpleCStarAlgebra_limit</code> Immediate Card prerequisites: Root compression becomes scalar in norm Used by in this scope: An irreducible CAR representation has no nonzero compact image Open Declaration Card Read exact source

Level 4 (1 Card)

Level 4 An irreducible CAR representation has no nonzero compact image Simplicity and infinite dimensionality rule out compact operators coming from nonzero CAR elements. MathlibAnnex.CStarAlgebra.CAR.eq_zero_of_isCompactOperator_image
Immediate Card prerequisites: Simplicity of the completed CAR algebra · The completed CAR algebra is infinite-dimensional Used by in this scope: Lifting a corner involution to an ambient CAR unitary · Realizing a finite Gram matrix in a represented CAR corner
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Level 5 (2 Cards)

Level 5 Lifting a corner involution to an ambient CAR unitary A prescribed unitary involution on a represented root corner can be realized on finitely many vectors by one lifted corner exponential. MathlibAnnex.CStarAlgebra.CAR.exists_liftedCornerExponential_apply_eq_involution
Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image Used by in this scope: Exact vector transport along an almost central unitary path · Approximating another vector state along a unitary path
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Level 5 Realizing a finite Gram matrix in a represented CAR corner The infinite-dimensional root corner contains an isometric copy of any finite family of vectors. MathlibAnnex.CStarAlgebra.CAR.exists_rootCornerFamily_with_gram
Immediate Card prerequisites: An irreducible CAR representation has no nonzero compact image Used by in this scope: Purifying a vector state on a finite CAR stage
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Level 6 (2 Cards)

Level 6 Purifying a vector state on a finite CAR stage Every finite-stage restriction of a unit vector state can be realized by a unit vector in any irreducible CAR representation. MathlibAnnex.CStarAlgebra.CAR.exists_unitVector_vectorFunctional_eq_on_stage
Immediate Card prerequisites: Realizing a finite Gram matrix in a represented CAR corner Used by in this scope: Approximating another vector state along a unitary path · Cross-representation state approximation with a protected finite set
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Level 6 Exact vector transport along an almost central unitary path Finite matrix-state tests ensure exact transport while protecting a prescribed finite set throughout the path. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_exact_unitary_path_apply_eq_and_conjugate_sub_norm_lt
Immediate Card prerequisites: Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: Cross-representation state approximation with a protected finite set
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Level 7 (2 Cards)

Level 7 Approximating another vector state along a unitary path A fixed irreducible CAR representation can approximate any vector state on finitely many elements by moving a given unit vector. MathlibAnnex.CStarAlgebra.CAR.exists_unitary_crossRepresentation_path_approx
Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Lifting a corner involution to an ambient CAR unitary · A finite-row average centralizes its matrix stage Used by in this scope: None in this selected scope
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Level 7 Cross-representation state approximation with a protected finite set Finite entrywise tests allow new state approximation while an entire unitary path nearly fixes earlier data. MathlibAnnex.CStarAlgebra.CAR.exists_stageTests_crossRepresentation_path_approx
Immediate Card prerequisites: Purifying a vector state on a finite CAR stage · Exact vector transport along an almost central unitary path Used by in this scope: None in this selected scope
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